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## Practical No. 1

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1.1.1. Calculate Momentum

24.01

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum  $p$  is calculated using the formula:  
$$p = m \times v$$
  
where:  
 $m$  is the mass of the object (in kilograms).  
 $v$  is the velocity of the object (in meters per second).  
**Input Format:**

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

**Output Format:**

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

Sample Test Cases

calculate...

Submit

1  
2  
3  
4

```
m=float(input())  
v=float(input())  
p=m*v  
print(f"{p:.2f}kgm/s")
```

Average time  
0.006 s  
5.75 ms

Maximum time  
0.007 s  
7.00 ms

2 out of 2 shown test case(s) passed  
2 out of 2 hidden test case(s) passed

Test case 1  
Expected output  
5.0  
10.0  
50.00kgm/s

Actual output  
5.0  
10.0  
50.00kgm/s

Test case 2

Terminal

Test cases

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1.1.2. Conditional Calculation Based on the Number of Digits

23.22

Write a Python program that accepts an integer  $n$  as input. Depending on the number of digits in  $n$ .  
**Constraints:**  
 $1 \leq n \leq 999$   
**Input Format:**

- The input consists of a single integer  $n$ .

**Output Format:**

- If  $n$  is a single-digit number, print its square.
- If  $n$  is a two-digit number, print its square root (rounded to two decimal places).
- If  $n$  is a three-digit number, print its cube root (rounded to two decimal places).
- Else print "Invalid".

Sample Test Cases

condition...

Submit

1  
2  
3  
4  
5  
6  
7  
8  
9

```
n = int(input())  
if 0<n<10:  
    print(f"{n**2}")  
elif 10<=n<100:  
    print(f"{n**0.5:.2f}")  
elif 100<=n<1000:  
    print(f"{n**(1/3):.2f}")  
else:  
    print("Invalid")
```

Average time  
0.005 s  
4.57 ms

Maximum time  
0.006 s  
6.00 ms

4 out of 4 shown test case(s) passed  
3 out of 3 hidden test case(s) passed

Test case 1  
Expected output  
9  
81

Actual output  
9  
81

Test case 2

Test case 3

Terminal

Test cases

Prev

Reset

Submit

Next

Course

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1.1.3. Days between Two Dates

31.13

Write a Python program to calculate number of days between two dates.

Input Format:

- The first line contains the first date in the format `YYYY - MM - DD`.
- The second line contains the second date in the format `YYYY - MM - DD`.

Output Format:

- An integer representing the number of days between the given dates.

Note:

- The first date should always be considered earlier than the second date.

Sample Test Cases

noOfDays...

Submit

Debugger

1from datetime import datetime

2

3

4

5d1 = datetime.strptime(input(), "%Y-%m-%d")

6d2 = datetime.strptime(input(), "%Y-%m-%d")

7

8print((d2 - d1).days)

Average time0.010 s10.25 msMaximum time0.027 s27.00 ms

2 out of 2 shown test case(s) passed

2 out of 2 hidden test case(s) passed

Test case 127 msDebug

Expected output

Actual output

2014-07-022014-07-02

2014-11-022014-11-02

123123

Test case 24 ms

TerminalTest cases

PrevResetSubmitNext

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1.1.4. Reverse a Number

07.63

Write a Python program to reverse the digits of the number and print the reversed number.

Input Format:

- The input is a single integer.

Output Format:

- The output prints a single integer, which is the reversed number.

Sample Test Cases

reverseN...

Submit

Debugger

1number = int(input())

2reverse\_number = int(str(number)[::-1])

3print(reverse\_number)

Average time0.005 s5.00 msMaximum time0.008 s8.00 ms

2 out of 2 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 18 msDebug

Expected output

Actual output

53675367

76357635

Test case 24 ms

TerminalTest cases

PrevResetSubmitNext

Course

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1.1.5. Multiplication Table

100%

Write a Python program that takes an integer as input and prints the multiplication table for that integer from 1 to 10.

Input Format:

- The first line of input contains an integer that represents the number for which the multiplication table is to be printed.

Output Format:

- Print the multiplication table for the given number in the format:  
`<number> x <multiplier> = <result>`

Note:

- The character "x" is used in the output to represent multiplication.
- Refer to the visible test-cases for more clarity.

Sample Test Cases

multipl...

Submit

Debugger

1n = int(input())2for i in range(1,11):3 print(f"{n} x {i} = {n\*i}")

Average time0.004 s4.00 msMaximum time0.005 s5.00 ms2 out of 2 shown test case(s) passed2 out of 2 hidden test case(s) passed

Test case 15 msDebug

Expected outputActual output

88

8 x 1 = 88 x 1 = 8

8 x 2 = 168 x 2 = 16

8 x 3 = 248 x 3 = 24

8 x 4 = 328 x 4 = 32

8 x 5 = 408 x 5 = 40

TerminalTest cases

PrevResetSubmitNext

Course

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1.2.1. Pass or Fail

100%

Write a Python program that accepts the number of courses and the marks of a student in those courses.

The grade is determined based on the aggregate percentage:

- If the aggregate percentage is greater than 75, the grade is Distinction.
- If the aggregate percentage is greater than or equal to 60 but less than 75, the grade is First Division.
- If the aggregate percentage is greater than or equal to 50 but less than 60, the grade is Second Division.
- If the aggregate percentage is greater than or equal to 40 but less than 50, the grade is Third Division.

Input Format:

- The first input will be an integer  $n$ , the number of courses.
- The second input will be  $n$  integers representing the marks of the student in each of the  $n$  courses, separated by a space.

Output Format:

- If the student passes all courses:
  - Print the aggregate percentage (formatted to two decimal places).
  - Print the grade based on the aggregate percentage.
- If the student fails any course (marks < 40 in any course), print:
  - "Fail".

Note:

- Aggregate Percentage: Average performance across all courses, calculated by summing all marks and dividing by the number of courses

passorFa...

Submit

Debugger

1subject=int(input())2marks=list(map(int,input().split()))3if any(mark<40 for mark in marks):4 print("Fail")5else:6 total\_marks=sum(marks)7 Aggregate\_percentage = (total\_marks/(subject\*100))\*1008 print(f"Aggregate Percentage:{Aggregate\_percentage:.2f}")9 if Aggregate\_percentage > 75:10 print("Grade: Distinction")11 elif Aggregate\_percentage > 60:

Average time0.005 s5.20 msMaximum time0.007 s7.00 ms5 out of 5 shown test case(s) passed5 out of 5 hidden test case(s) passed

Test case 17 msDebug

Expected outputActual output

44

55 65 78 8955 65 78 89

Aggregate Percentage: 71.75Aggregate Percentage: 71.75

Grade: First DivisionGrade: First Division

Test case 24 ms

TerminalTest cases

PrevResetSubmitNext

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1.2.2. Fibonacci Series

Write a Python program that uses recursion to print the first  $n$  terms of the Fibonacci series.

Input Format:

- A single integer  $n$  representing the number of terms to generate.

Output Format:

- A single line containing the first  $n$  terms of the Fibonacci sequence, separated by spaces.

Sample Test Cases

fibonacci....

Submit

1# def fibonacci(n):  
2  
3# return fibonacci()  
4  
5  
6n = int(input())  
7# for i in range(1, n + 1):  
8# print(fibonacci(i), end=" ")  
9  
10  
11def fib(n):  
12if n == 0:  
13return 0  
14if n == 1:  
15return 1  
16return fib(n-1) + fib(n-2)

Average time: 0.011 s  
Maximum time: 0.028 s  
2 out of 2 shown test case(s) passed  
2 out of 2 hidden test case(s) passed

Test case 1 5 ms Debug  
Expected output: 0 1 1 2 3  
Actual output: 0 1 1 2 3

Test case 2 4 ms

Terminal Test cases

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1.2.3. Pattern - 1

Write a Python program to print a pattern of asterisks in the form of a right-angled triangle.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format

- The output should display the pattern of asterisks (\*), with each row containing an increasing number of asterisks.

Note: Refer to the displayed test cases for the sample pattern.

Sample Test Cases

rightangl...

Submit

Debugger

1 num = int(input())  
2 for i in range(1,num+1):  
3 print("\*"\*i)

Average time: 0.004 s  
Maximum time: 0.008 s  
3.67 ms 8.00 ms

2 out of 2 shown test case(s) passed  
4 out of 4 hidden test case(s) passed

Test case 1 3 ms Debug

Expected output: 5  
\*  
\* \*  
\* \* \*  
\* \* \* \*  
\* \* \* \* \*

Actual output: 5  
\*  
\* \*  
\* \* \*  
\* \* \* \*  
\* \* \* \* \*

Terminal Test cases

Prev Reset Submit Next

Course

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1.2.4. Pattern - 2

Write a Python program to print a right-angled triangle pattern of numbers.

Input Format:

- The input is an integer, representing the number of rows in the pattern.

Output Format:

- The output should display the pattern of numbers separated by space, with each row containing increasing numbers starting from 1 up to the row number.

Note:

- Refer to the displayed test cases for the sample pattern.

Sample Test Cases

numberP...

Submit

Debugger

1 # Type Content here...  
2 n = int(input())  
3 for i in range(1,n+1):  
4 for j in range(1,i+1):  
5 print(j,end=' ')  
6 print()  
7

Average time: 0.004 s  
Maximum time: 0.005 s  
4.26 ms 5.00 ms

2 out of 2 shown test case(s) passed  
2 out of 2 hidden test case(s) passed

Test case 1 4 ms Debug

Expected output: 5  
1  
1 2  
1 2 3  
1 2 3 4  
1 2 3 4 5

Actual output: 5  
1  
1 2  
1 2 3  
1 2 3 4  
1 2 3 4 5

Terminal Test cases

Prev Reset Submit Next